Particle PhysicsNotes

Vyas Mokal

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Contents

Chapter 1

Lecture 4

1.1 Language of Particle Physics

- Theoretical/Mathematical $\rightarrow$ QFT $\rightarrow$ Symmetries
- Numerical/Computation
- Data Handling [1]

1.2 Units of Particle Physics

- Quantum Mechanics in SI[2] $\rightarrow$ lots of $\hbar$ and c
- adapt a unit system where $\hbar = c = 1$
  This takes care of not carrying these extra factors

1.2.1 Outcomes of assuming this

- Velocity in special relativity $\beta = \frac{v}{c}$ becomes dimensionless
- Action is dimensionless $\rightarrow$ Energy $\times$ Time
- Mass dimension is same as energy dimension

$$[\text{mass}] = [\text{Energy}] = [\text{momentum}] \xrightarrow{\text{in}} \text{GeV}$$

$$E^2 = p^2c^2 = m^2c^4 \rightarrow E^2 = p^2 + m^2$$

[1] in LHC large data 1 Trillion bytes per second is generated
[2] Standard Units

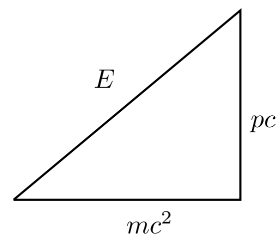

Fig. 4.1

- [Length] = [Time] = $[\text{Energy}]^{-1}$
  To see this use $c = \frac{\text{Length}}{\text{Time}}$, now since velocity is dimensionless $\implies$ Length and Time have same dimensions.
  Use $\Delta E \Delta t \sim \hbar$, since $\hbar$ is dimensionless Energy will have dimension inverse of time.

- Consider the following table:

Quantity	SI	NU	Unit
Energy	ML^2T^{-2}	1	GeV
Momentum	MLT^{-1}	1	GeV
Time	T	-1	GeV^{-1}
Length	L	-1	GeV^{-1}
Cross Section(σ)	L^2	-2	GeV^{-2}
Action	ML^2T^{-1}	0	dimensionless
Electric Charge	IT	0	dimensionless

  Here NU: Natural Units
  Cross section is conventionally measured in barns 1 barn = $1 \times 10^{-28} m^2$ and need convert accordingly.

- To interpret the result calculated using NU we need to convert it to SI units.

- insert $\hbar$ and c untill the dimensions match back